

Paper Practice 1.2**Algebraic Expressions**

Show all of your work in the space provided. Define your variables when a problem asks you to. Circle your final answers.

- 1.** Write a verbal expression for each algebraic expression. (Translate)

a. $7x^3 + 4$

b. $(3/4)n^2$

c. $(m^5 + 2p) \div 9$

- 2.** Write an algebraic expression for each verbal expression. (Translate)

a. three times the quantity k plus 8

b. seven less than a number squared

c. twenty increased by the product of five and d

- 3.** Evaluate $3x^2 - 2y$, if $x = 4$ and $y = 5$. (Evaluate)

- 4.** Evaluate $b(12 - c)/a^2$, if $a = 2$, $b = 5$, and $c = 4$. (Evaluate)

- 5.** Evaluate $a^3(4b - a + 1) - c$, if $a = 2$, $b = 3$, and $c = 6$. (Evaluate)

- 6.** Evaluate $b(7 - c)/a^2$, if $a = 3$, $b = 9$, and $c = 3$. (Evaluate)

- 7.** Identify the terms, the coefficients, and the constant in $6x^2 + 4x - 9$. (Vocabulary)

- 8.** In a soccer tournament a team earns 3 points for a win and 1 point for a tie. Define your variables, then write an expression for a team's total points. (Write)

- 9.** Using your expression from #8, find the points for a team with 5 wins and 2 ties. (Evaluate)

- 10.** Error Analysis – Gus evaluated $2x^2$ when $x = 3$ and wrote 36. Identify his mistake, then evaluate correctly. (Error Analysis)

- 11.** Write a verbal expression for $a^4 + 6b/7$, then write one for $(a^4 + 6b) \div 7$. Explain how the wording shows the difference. (Compare)

- 12.** The surface area of a sphere is four times π times the radius squared. Write the expression, then find the surface area of a bubble with radius 12 cm. Round to the nearest tenth. (Write & Evaluate)